Spherical Cyclic Time v2.0 Addendum
Minimal Closed-Background Transfer Tests

Public synthesis addendum

2026-07-04

Abstract

This addendum summarizes the v2.0-A through v2.0-F branch of Spherical Cyclic Time (SCT). The branch was opened after an external referee-style review argued that v1.7-v1.9 should be treated as requirements and obstruction audits, not as computational negative results. The v2.0 branch therefore refocused on a minimal computational question: whether a preregistered closed toy background can produce a perturbative transfer shape compatible with the Gaussian/diffusive ansatz $\exp[-(k/k_c)^2]$. Two independently preregistered toy background families were tested. Both produced bounded, high-k plateau-like transfer shapes and were classified as non-Gaussian transfer under the locked rules. The result is a limited computational negative result for the tested closed-background route. It is not a no-go theorem for cyclic or bounce cosmology and it is not a physical cosmology or observational claim.

1 Purpose of this addendum

The previous v1.7-v1.9 line of work mapped requirements and obstructions for turning SCT from an informational toy framework into a more physical open-system or background-transition model. That line was useful as governance, but it did not produce a new computational result. The v2.0 branch was therefore deliberately narrowed to a single computable target:

$$T_R(k) \quad \text{from a closed, preregistered toy background.}$$

The branch did not attempt to derive a bath, a noise kernel, P_ξ, Γ_S, FDT/KMS conditions, an entropy map, or an observational fit. The only tested question was whether the closed-background transfer shape resembles the Gaussian/diffusive form used as a conditional toy ansatz in earlier versions.

2 Locked minimal protocol

The proxy perturbation equation was fixed before execution:

$$u_k'' + \left(k^2 - \frac{a''}{a}\right) u_k = 0.$$

The mode grid was fixed as

$$k = \mathrm{logspace}(-2, 2, 121), \qquad y \in [-50, 50].$$

The transfer shape was extracted using the preregistered two-basis norm protocol. The Gaussian comparison

$$T_G(k) = \exp[-(k/k_c)^2]$$

was used only as a diagnostic comparator, not as a target for tuning.

3 Background families tested

Two background families were preregistered and then executed.

Family 1: smooth global power bounce.

$$a(y) = (1 + y^2)^{\alpha/2}.$$

Primary value: $\alpha = 0.20$. Sensitivity values: $\alpha = 0.10$ and $\alpha = 0.50$.

Family 2: compact C^2 matched power-law bounce. For $|y| \geq 1$,

$$a(y) = |y|^{\alpha}.$$

For $|y| < 1$,

$$a(y) = a_0 + by^2 + cy^4,$$

with

$$c = \frac{\alpha(\alpha - 2)}{8}, \qquad b = \frac{\alpha(4 - \alpha)}{4}, \qquad a_0 = 1 - \frac{\alpha(6 - \alpha)}{8}.$$

The same primary and sensitivity values were used.

4 Results

All six executed cases were classified as non-Gaussian transfer under the locked rules. The primary results are shown below.

Background family	α	$T_{\min}$	high-k plateau	Gaussian R^2
Smooth global power bounce	0.20	0.496693	0.496709	-3.337
Compact C^2 matched bounce	0.20	0.484690	0.484705	-2.021

The sensitivity cases gave the same qualitative classification: non-Gaussian transfer with bounded/high-k plateau behavior.

Background family	α	classification	high-k plateau
Smooth global power bounce	0.10	non-Gaussian	0.834813
Smooth global power bounce	0.50	non-Gaussian	0.031437
Compact C^2 matched bounce	0.10	non-Gaussian	0.830421
Compact C^2 matched bounce	0.50	non-Gaussian	0.022801

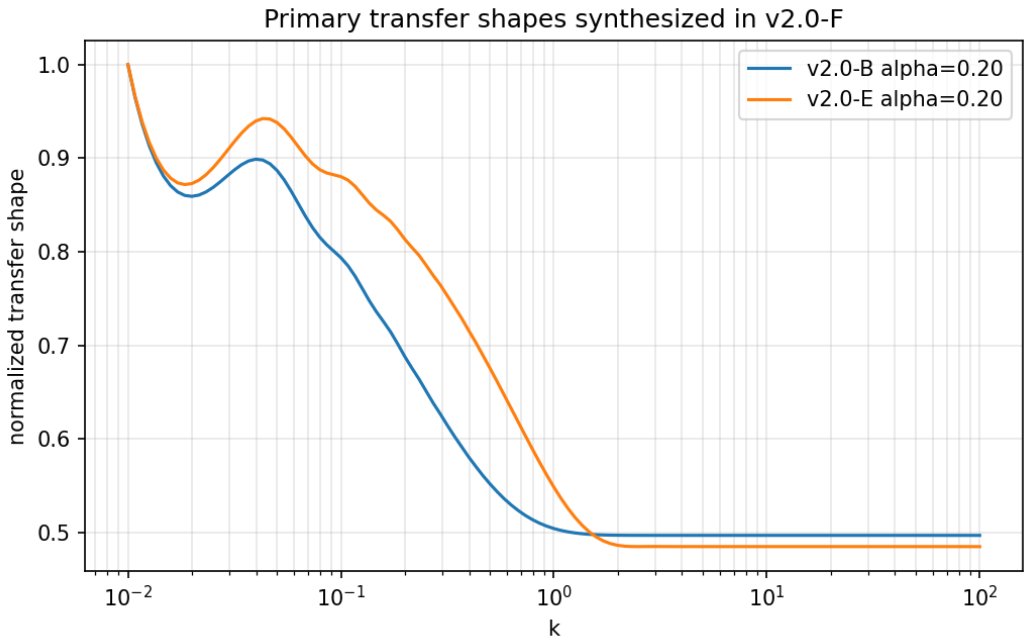


Figure 1: Primary transfer shapes synthesized in v2.0-F and included here for public readability. Both primary cases approach a bounded high-k plateau rather than an indefinitely decaying Gaussian exponential.

5 Interpretation

The two tested closed-background routes do not support the Gaussian/diffusive transfer ansatz as a derived output. The result is stronger than the v1.7-v1.9 obstruction survey because it comes from preregistered numerical execution, not only from requirements inspection.

The correct conclusion is limited:

> The Gaussian/diffusive ansatz $\exp[-(k/k_c)^2]$ is not supported by the two preregistered closed-background toy families tested in v2.0-B and v2.0-E.

The result should not be overstated. It does not falsify all cyclic or bounce backgrounds. It does not derive a physical kernel. It does not establish a general no-go theorem. It does not address open-system kernels, stochastic noise, entropy production, or a physical restart mechanism.

6 Claim boundary

Allowed public claims:

- v2.0 refocused SCT onto a minimal computational transfer test.
- Two toy background families were preregistered before execution.
- Both produced non-Gaussian, bounded/high-k plateau transfer shapes.
- The current minimal closed-background T_R route is closed as a limited negative computational result.

Forbidden claims:

- no general no-go theorem for cyclic or bounce cosmology;
- no observational claim;
- no physical cosmology claim;
- no derivation of P_ξ, Γ_S, FDT/KMS, bath physics, entropy map, or restart dynamics;
- no claim that SCT as an informational framework is falsified.

7 Release note

For public release, v2.0 should be presented as an addendum or synthesis package, not as a replacement for the full v1.6/v1.9 internal archive. A flat release package should include this addendum, the result tables, the execution scripts from v2.0-B and v2.0-E, and a hash ledger. Recursive internal recovery packages should remain internal.

Lineage anchors

This addendum summarizes the local artifacts v1.9-R and v2.0-A through v2.0-F. Hashes are recorded in the accompanying lineage hash ledger. This document is not peer reviewed.